

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder and Software Engineer with over 4 years of experience, adept at designing and implementing scalable microservices and distributed data pipelines using Java, Python, Spark, SQL, AWS, and Azure. Proven ability to optimize system performance, automate CI/CD processes, and leverage RDBMS and NoSQL databases for robust data solutions. Expertise extends to infrastructure automation with Terraform and developing privacy-preserving AI applications, aligning with core requirements for a Java Software Engineer role focused on system architecture and performance.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP APIs using Java 8/11, Spring Boot, and Azure Functions, managing over 50,000 daily requests with 99.99 percent uptime and adhering to AMQP standards.
- Automated CI/CD pipelines via Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments, integrating unit tests for quality assurance.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling to reduce issue resolution time by 20 percent, leveraging Object-Oriented Programming principles.
- Orchestrated a major EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, ensuring robust server-side logic.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed enterprise applications and RESTful web services using Java and EJB for automotive clients, focusing on system architecture and J2EE best practices.
- Achieved a 90 percent system performance improvement through SQL query optimization, database tuning of RDBMS, and modernization of legacy Java code.
- Implemented CI/CD pipelines utilizing Jenkins and GitLab, incorporating automated testing and quality gates for enhanced product development.
- Mentored junior developers through code reviews, debugging sessions, and promoting clean coding practices, fostering professional development within the team.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization of server-side logic.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity within relational database environments.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical application modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to effective product management.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, AWS, GCP, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline processing over 8 GB of NYC taxi trip data in near real-time, leveraging PySpark for efficient data manipulation and utilizing Kafka for streaming ingestion.
- Engineered an interactive Flask web application for data exploration with dynamic filtering and visualization, employing REST APIs for data access.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and efficient orchestration.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and improving distributed computing efficiency.
- Automated cloud infrastructure provisioning using Terraform across AWS (S3, RDS, ECS, Lambda, QuickSight) and GCP services, ensuring scalable data storage and management.

Privacy-Preserving AI Email Client (Capstone Project - Hugging Face, T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, integrating Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA for low-resource training under strict privacy constraints, exploring advanced machine learning techniques.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, demonstrating expertise in data analysis and virtualisation.
- Explored multi-threading for parallel processing of email data and investigated concepts related to NoSQL databases for potential

unstructured data storage.

SKILLS

- **Programming Languages:** Proficient in Java, Python, and SQL for developing robust applications and data pipelines. Experience with J2EE frameworks and server-side logic.
- **Frameworks & Libraries:** Skilled in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable microservices and AI models.
- **Web Services & Messaging:** Expertise in designing and implementing REST APIs, SOAP APIs, and integrating with messaging queues like AMQP and RabbitMQ for seamless communication.
- **Cloud Platforms:** Extensive experience with AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and GCP for cloud management and infrastructure automation.
- **Databases:** Proficient with RDBMS like PostgreSQL, Oracle Database, IBM DB2, and RDS, alongside experience with NoSQL databases for diverse data storage needs.
- **Tools & Technologies:** Adept in Docker, Kubernetes for orchestration, Jenkins for CI/ CD, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for infrastructure as code.
- **Concepts:** Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object- Oriented Programming (OOP), Microservices Architecture, Large- scale data platforms, System Architecture, Distributed Computing, Data Structure, Multi-Threading, and Networking.
- **Data Analysis & Visualization:** Experienced in data analysis and visualization using Tableau and QuickSight, enabling actionable insights and informed decision-making. Familiar with GRPC for efficient inter-service communication.